

Graph Algorithms: Shortest Path & Minimum Spanning Trees

Subject: CS301 | Unit 4: Advanced Graphs

1. Executive Overview & Syllabus Scope

Detailed pseudocode, time complexity proofs, and step-by-step traces for Dijkstra, Bellman-Ford, Floyd-Warshall, Kruskal, and

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 4: Advanced Graph Algorithms & Complexity Proofs
- 1. Graph Representations: Adjacency Matrix $O(V^2)$ vs Adjacency List $O(V+E)$ tradeoffs
- 2. Single-Source Shortest Path: Dijkstra's Algorithm with Min-Heap in $O(V \log V)$
- 3. Bellman-Ford Algorithm: Handling negative edge weights and cycle detection in $O(V \cdot E)$
- 4. All-Pairs Shortest Path: Floyd-Warshall Dynamic Programming in $O(V^3)$
- 5. Disjoint Set Union DSU: Path Compression and Union by Rank for near-linear $O(\alpha V)$
- 6. Minimum Spanning Trees MST: Kruskal's greedy edge sorting vs Prim's vertex expansion

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.